

Tensorflow model pruning

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Importance based pruning

Harvesting Hyperparameters, Kernels, and Matrices

This project the MNIST dataset for digit recognition was used

Motivation is to upgrade to a full OCR dataset

The requested harvesting was executed

Then several pruning techniques were applied to analyze what are the most accurate

Model: CNN with fully connected layers

Input Shape: (64, 64, 3)

Total Layers: 5

1 Convolutional Layer

1 MaxPooling Layer

1 Flatten Layer

2 Dense Layers (Fully Connected)

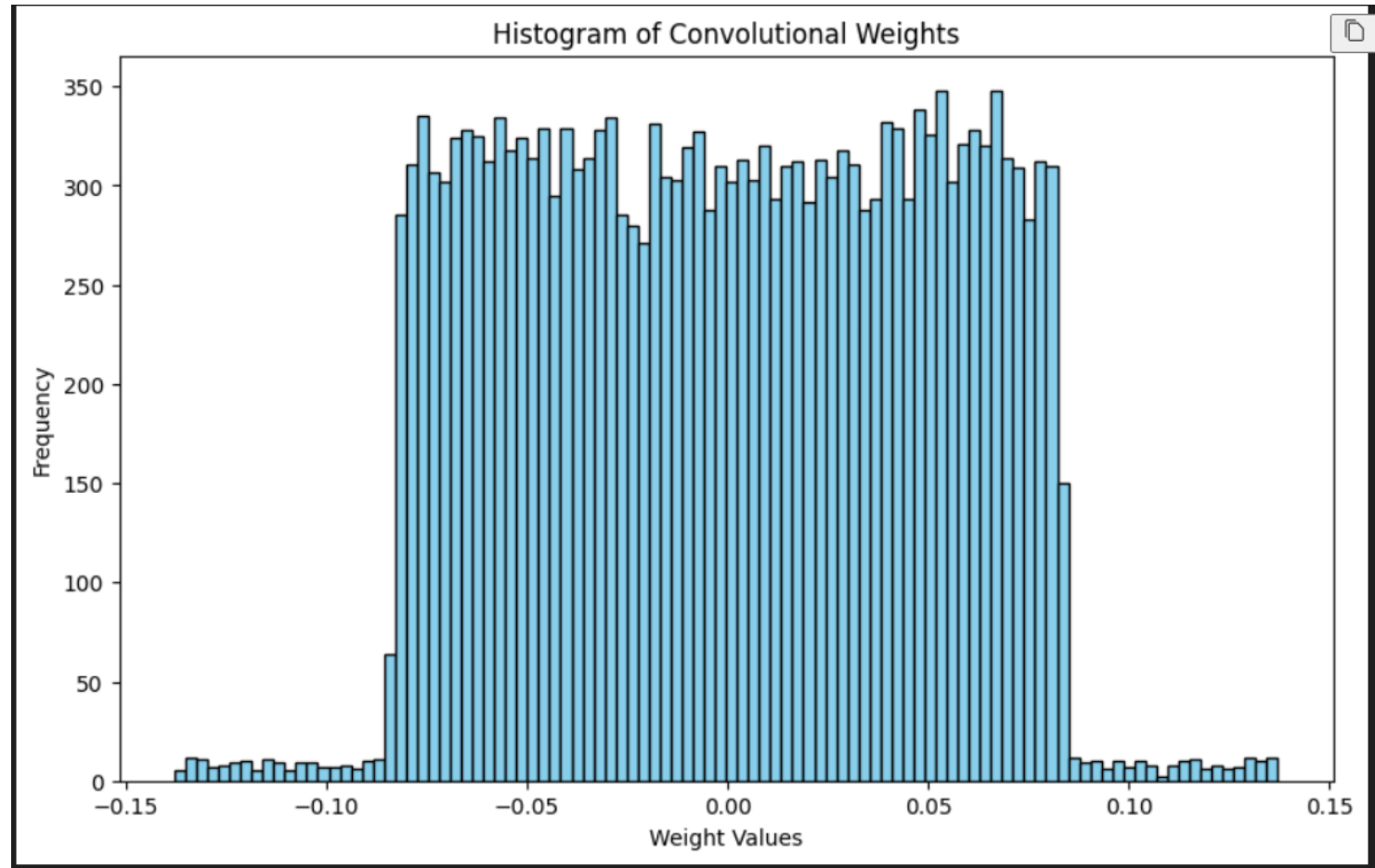
Key Parameters:

Conv Layer 1: 32 filters, kernel size (3, 3), ReLU activation

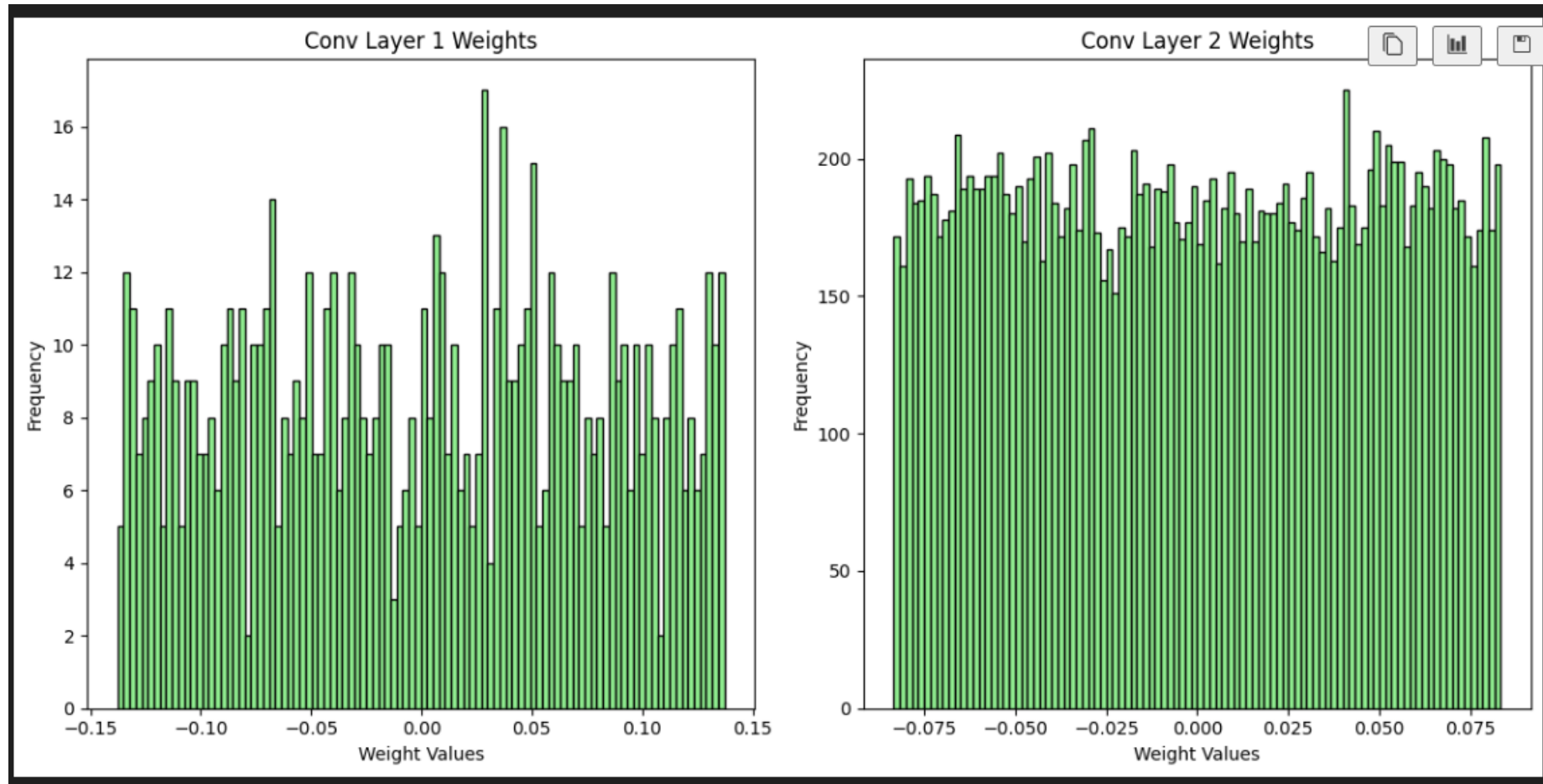
Dense Layer 1: 128 units, ReLU activation

Dense Layer 2: 10 units, Softmax activation for classification

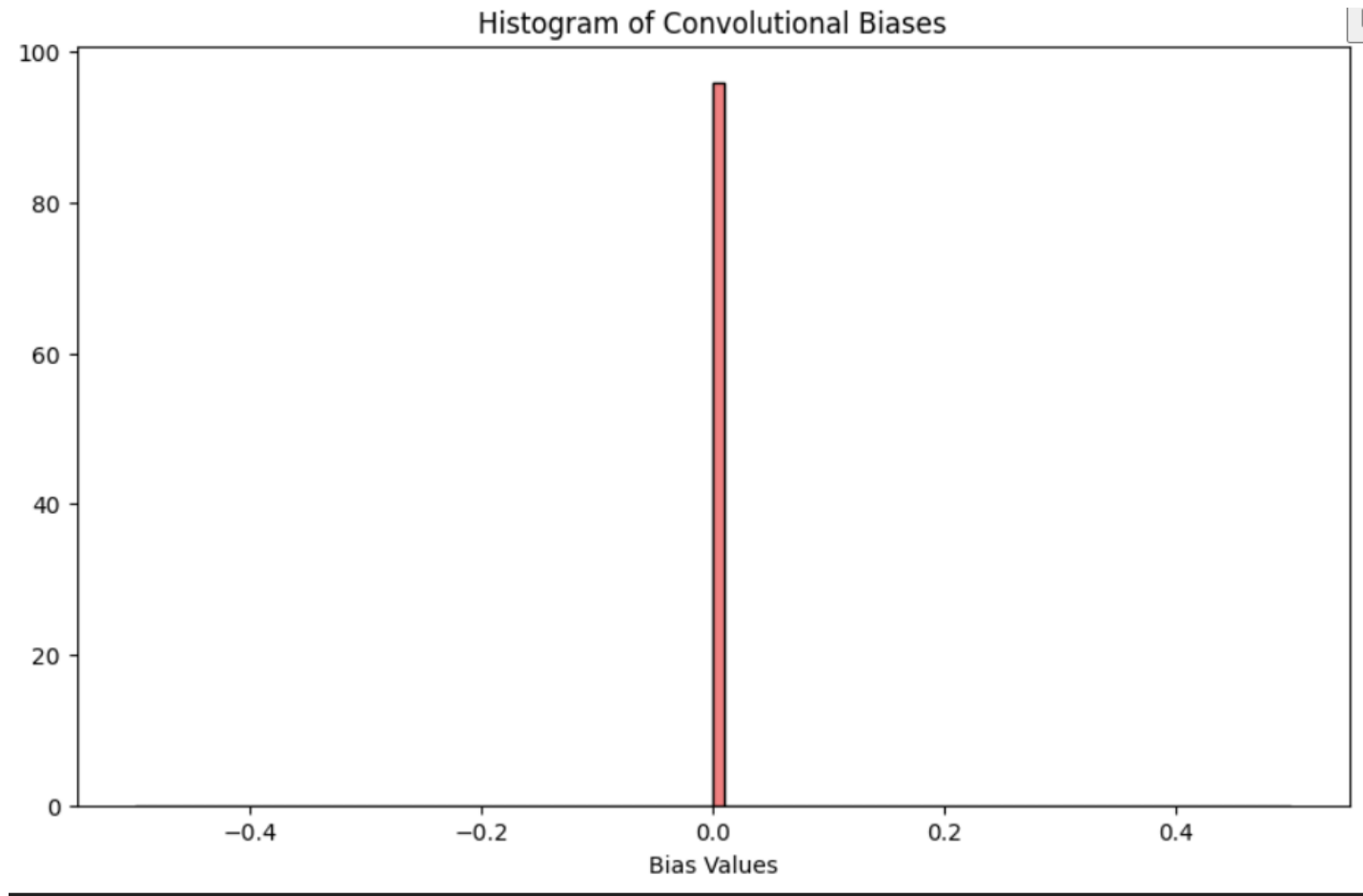
Histogram of Conv Weights



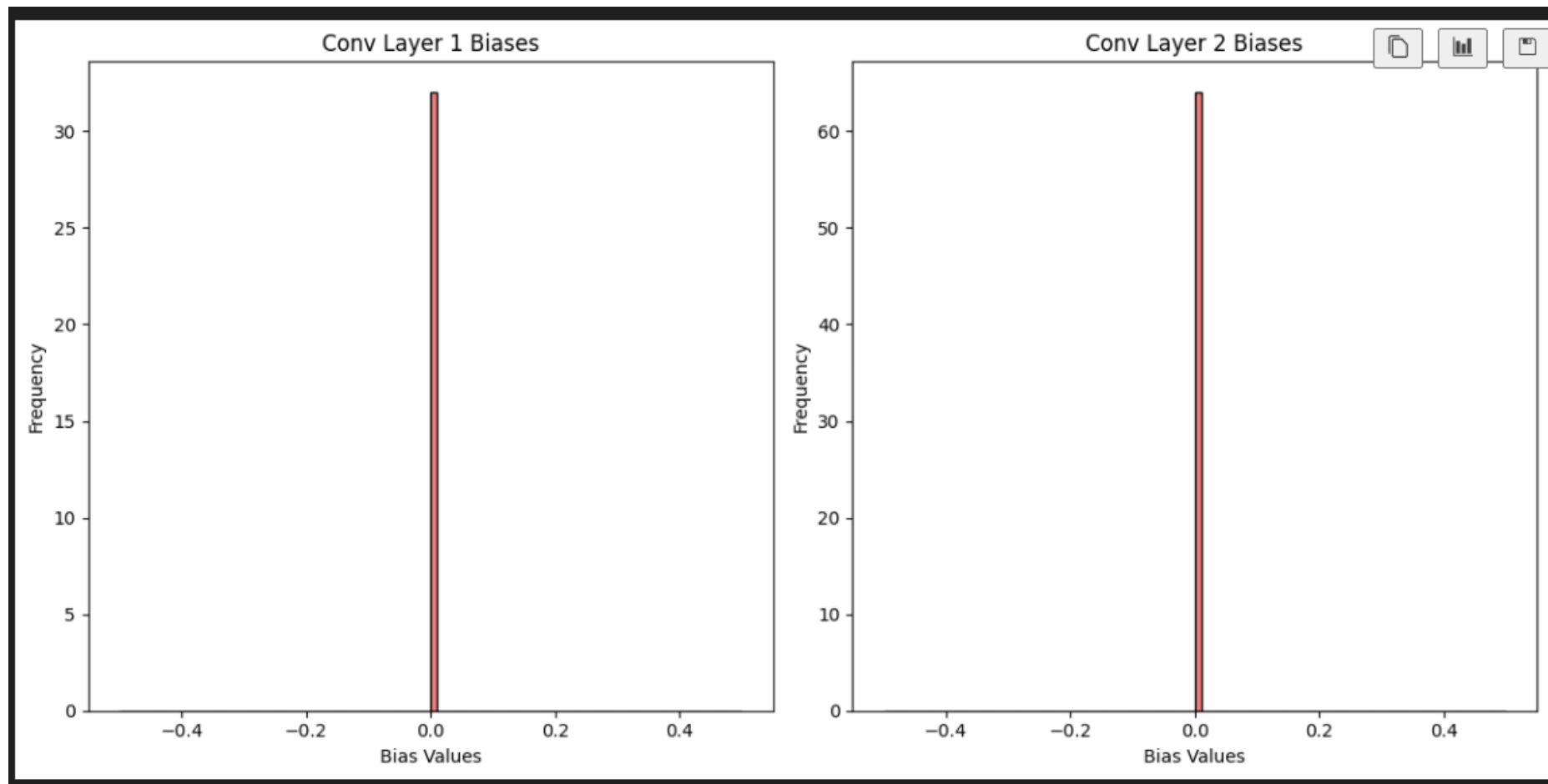
Histogram of Conv Weights Per Layer



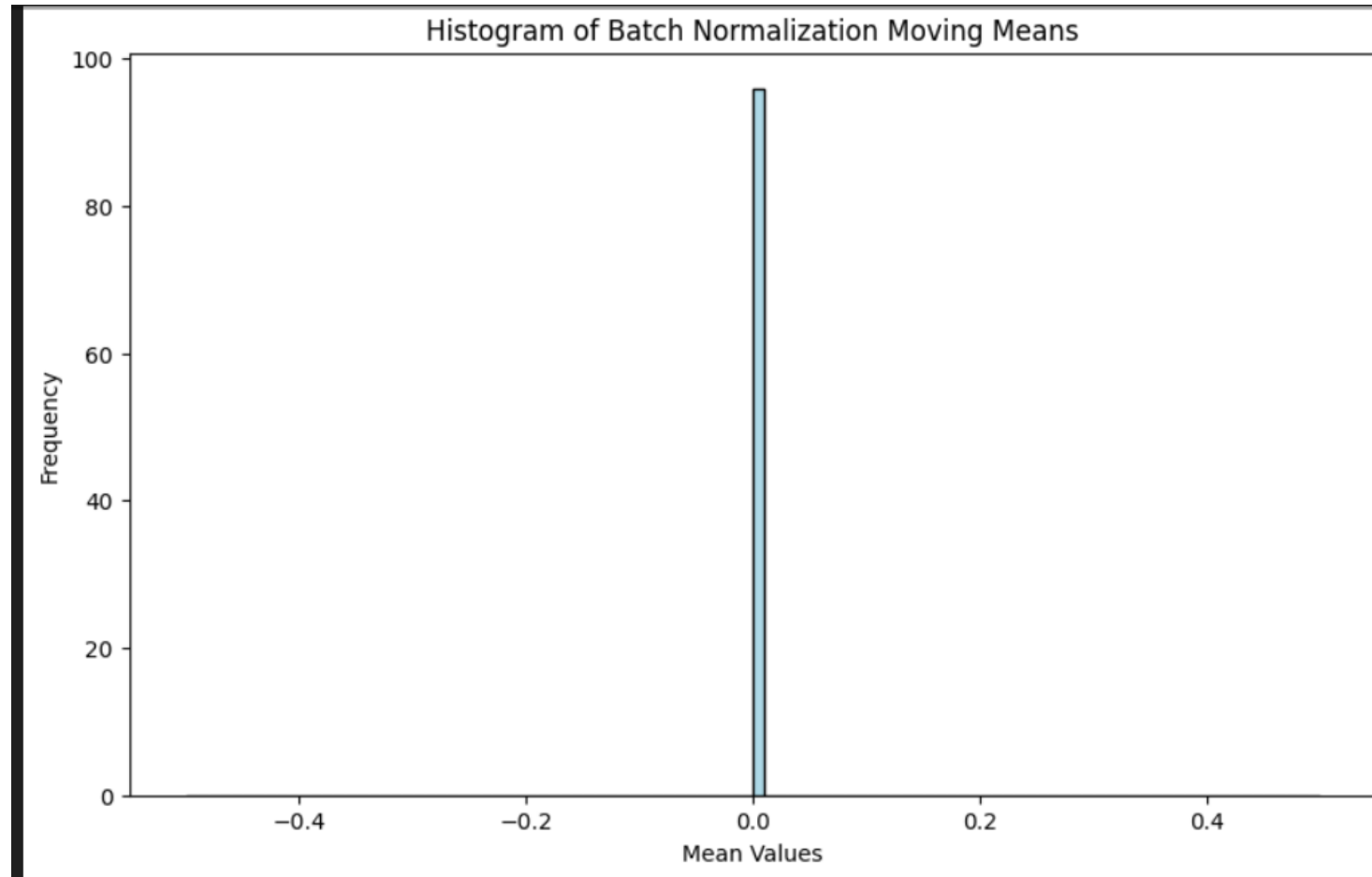
Convolutional bias(contrary to ResNet)



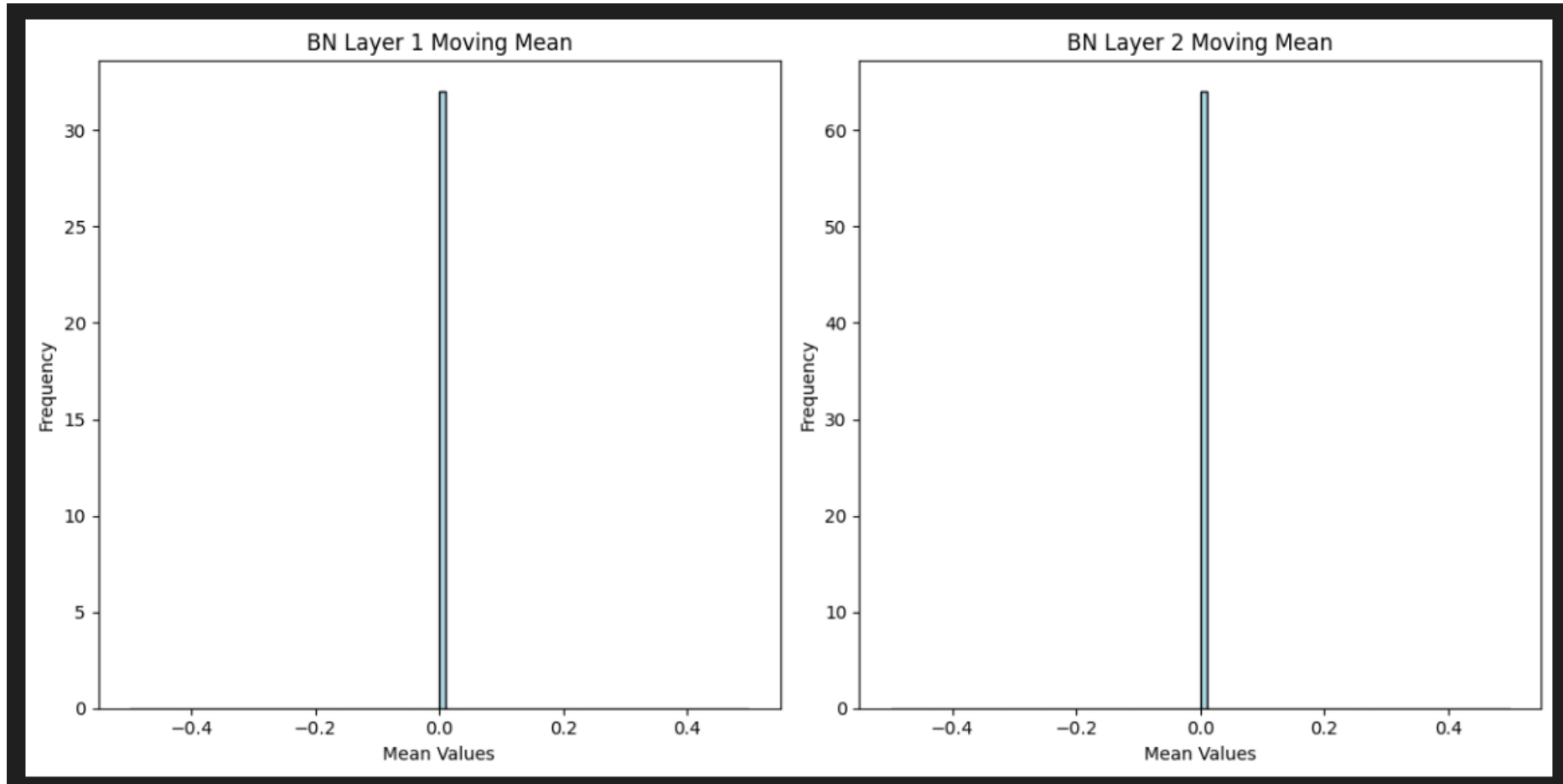
Convolutional bias



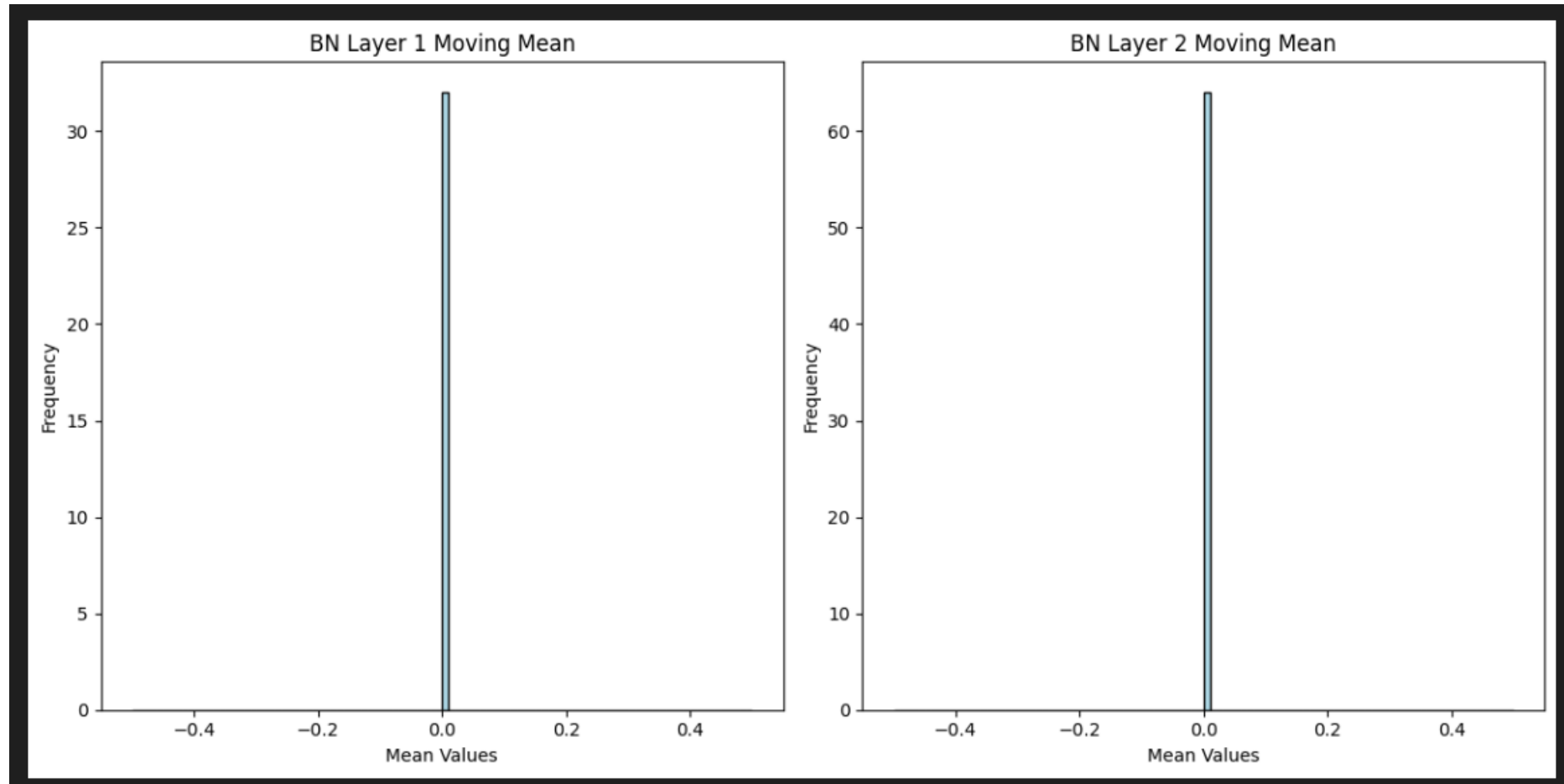
Batch Normalization Trained Mean (Variance)



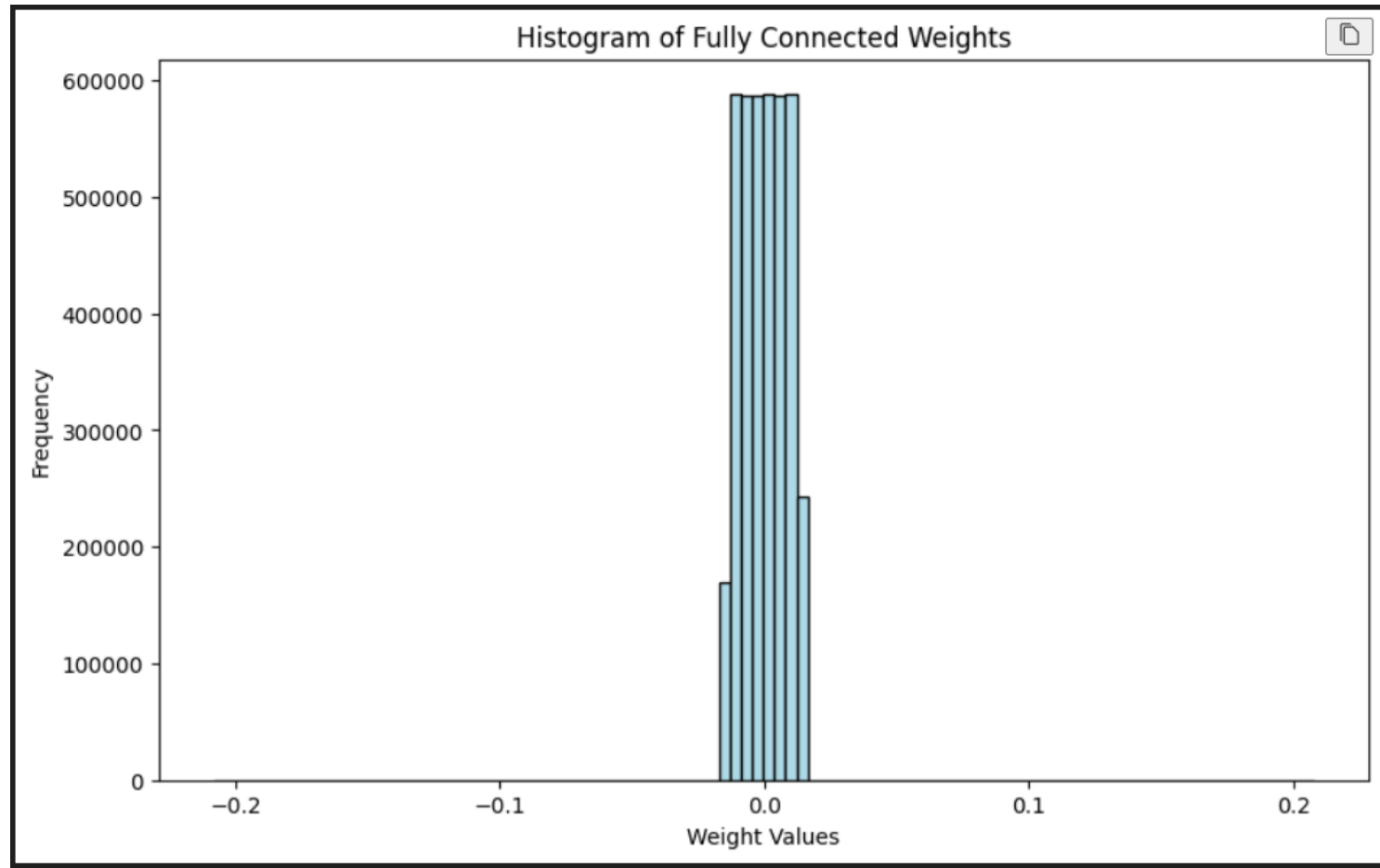
Batch Normalization Trained Mean (Variance)



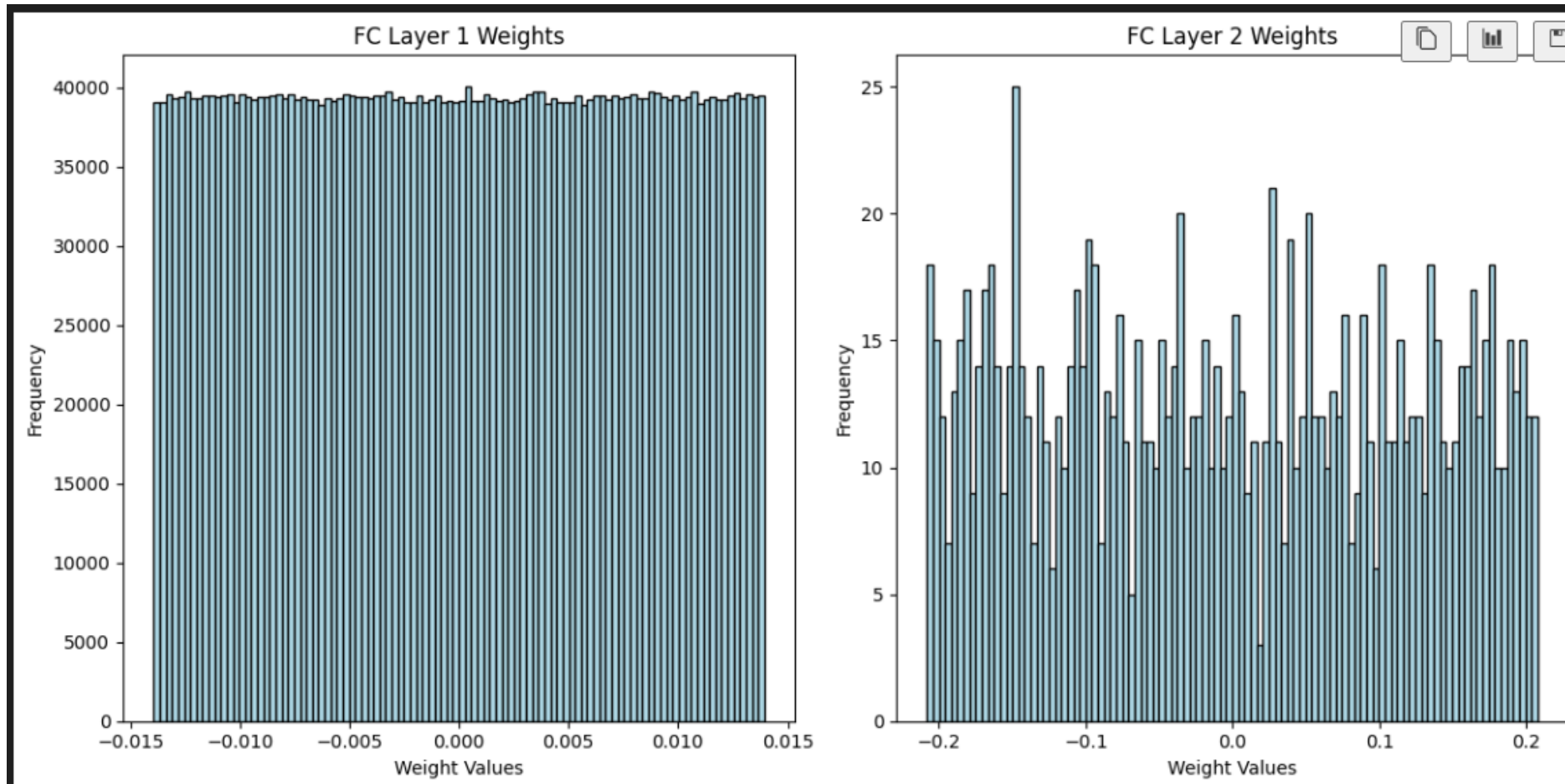
Batch Normalization Trained Variance



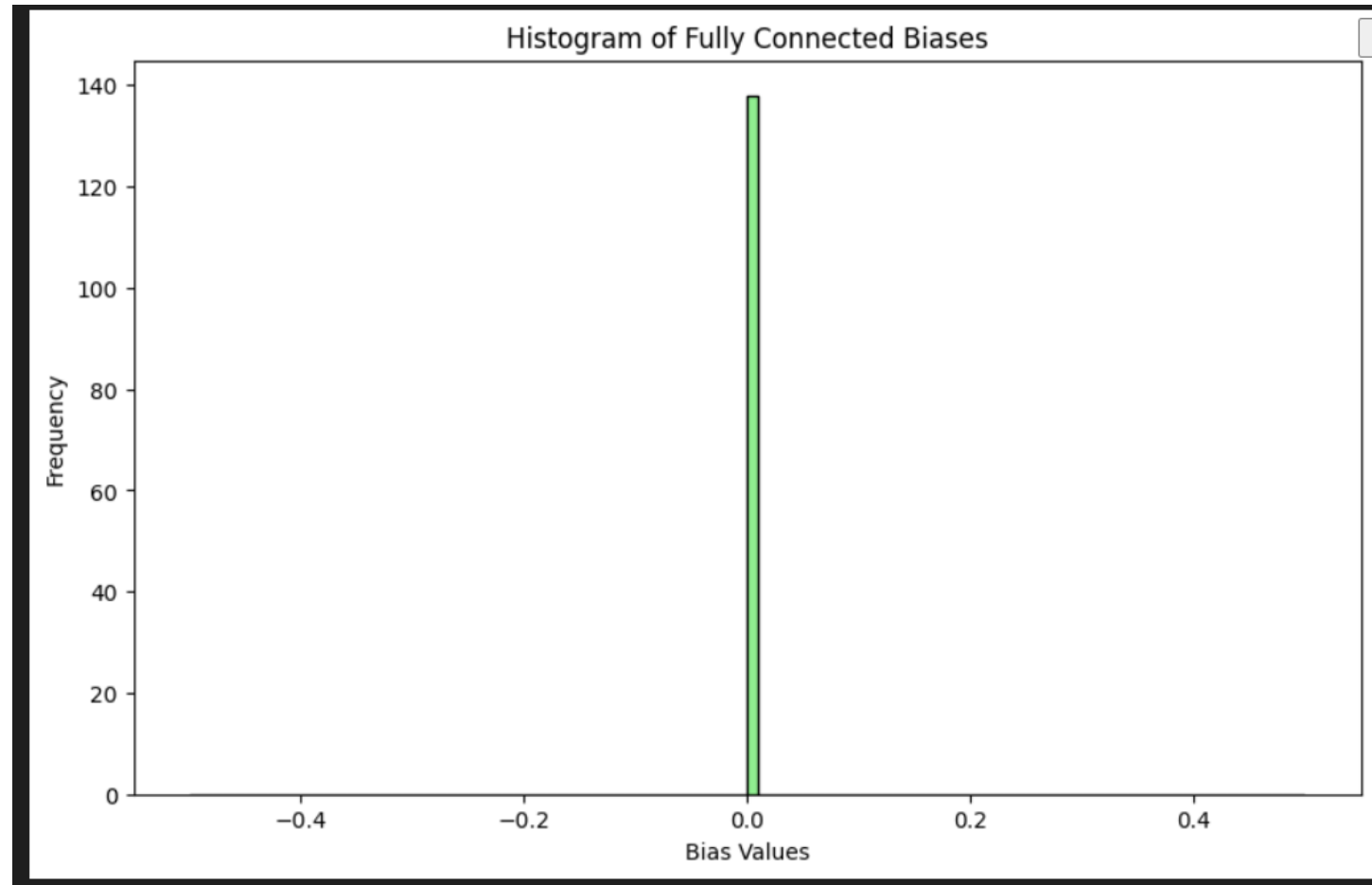
Fully connected weights



Fully connected weights



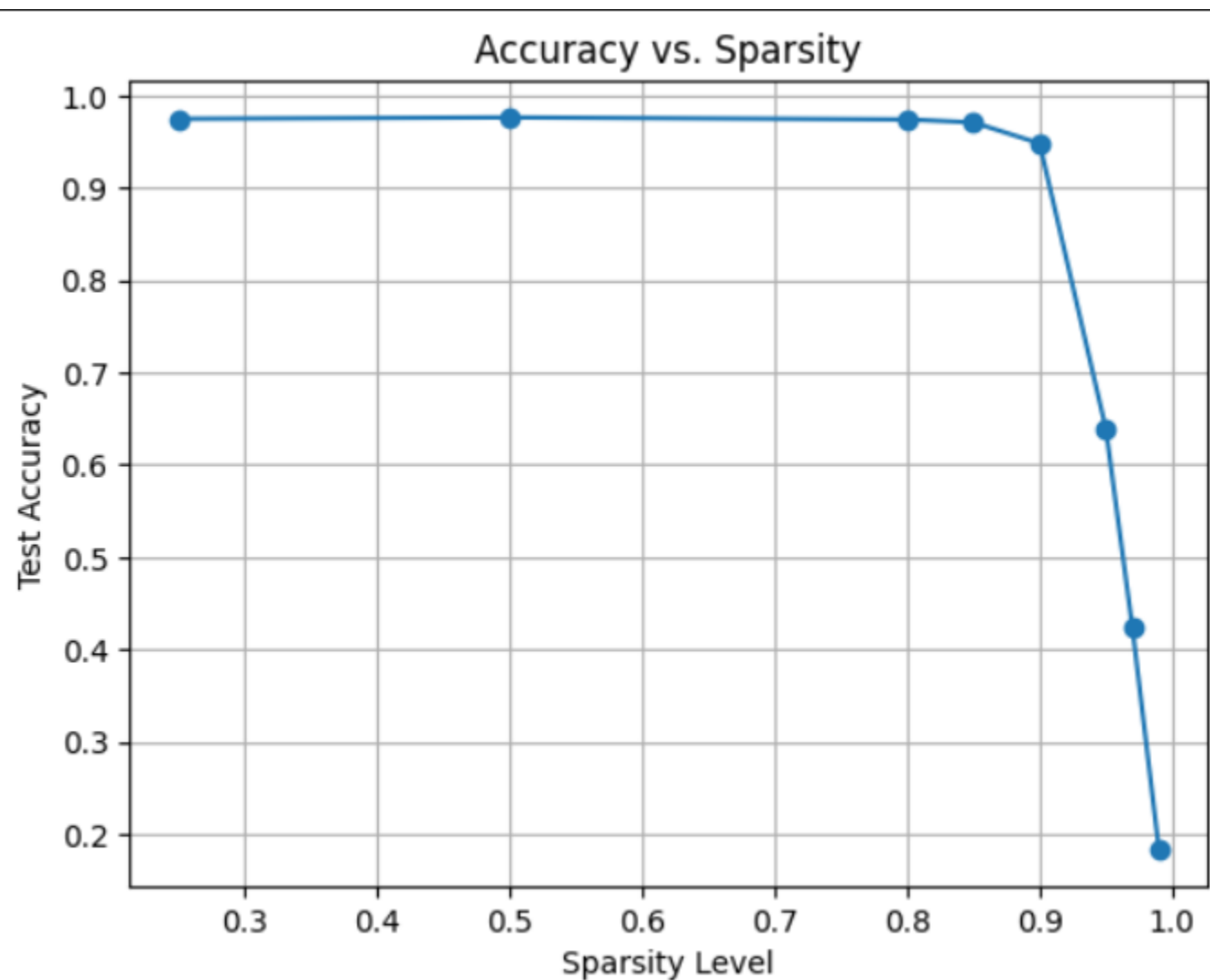
Fully connected Bias



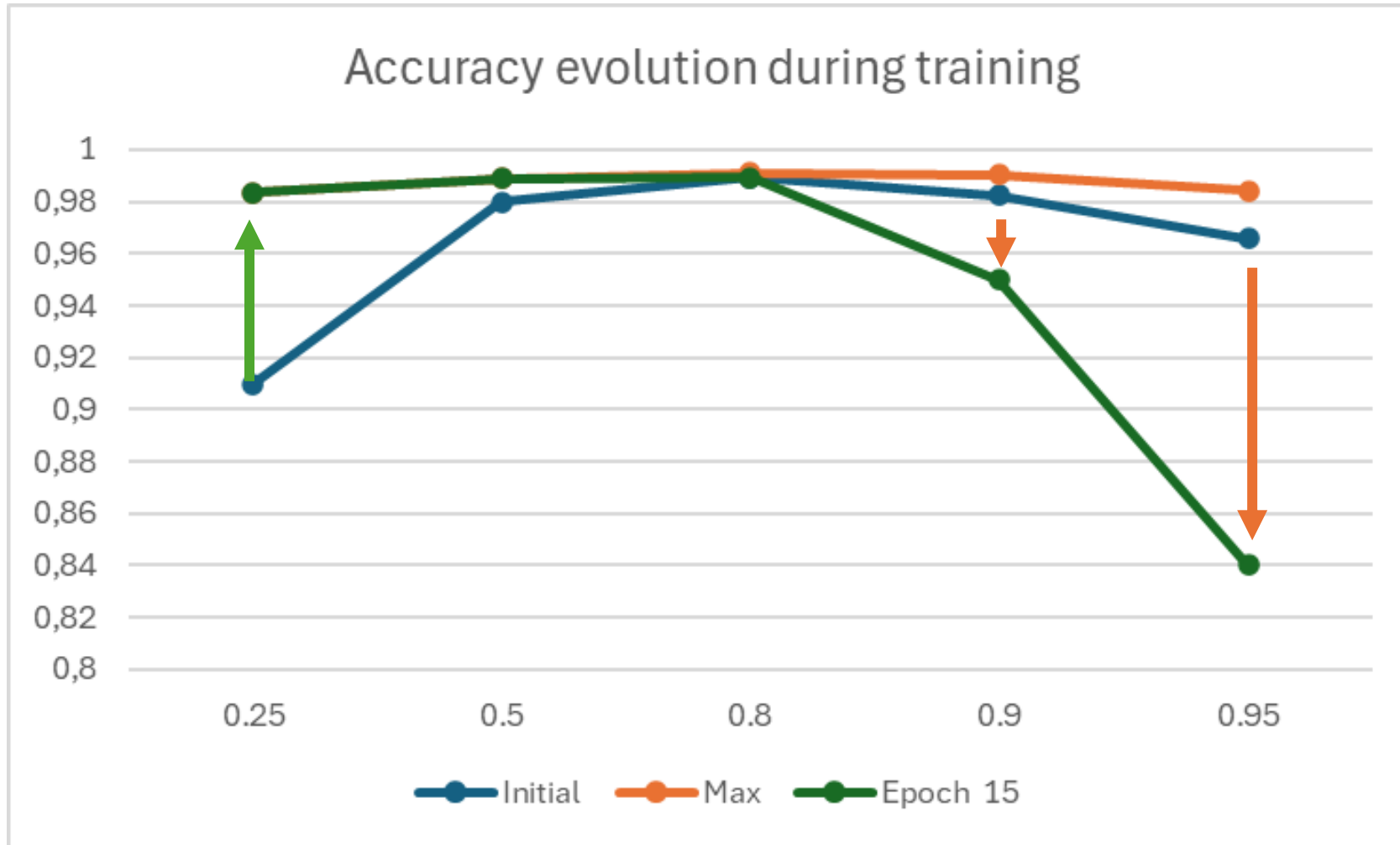
Model training and its accuracy



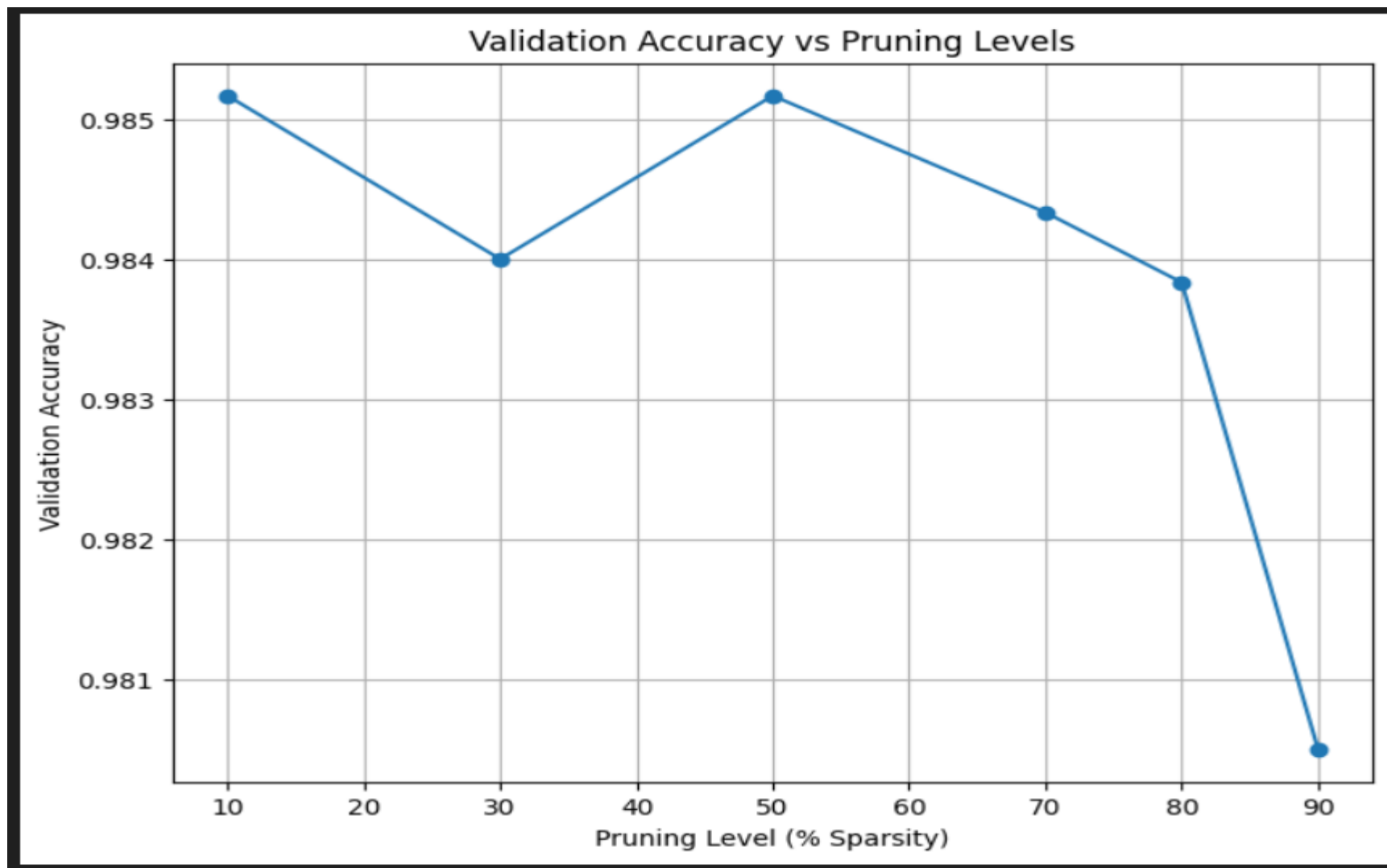
Magnitude-Based Pruning



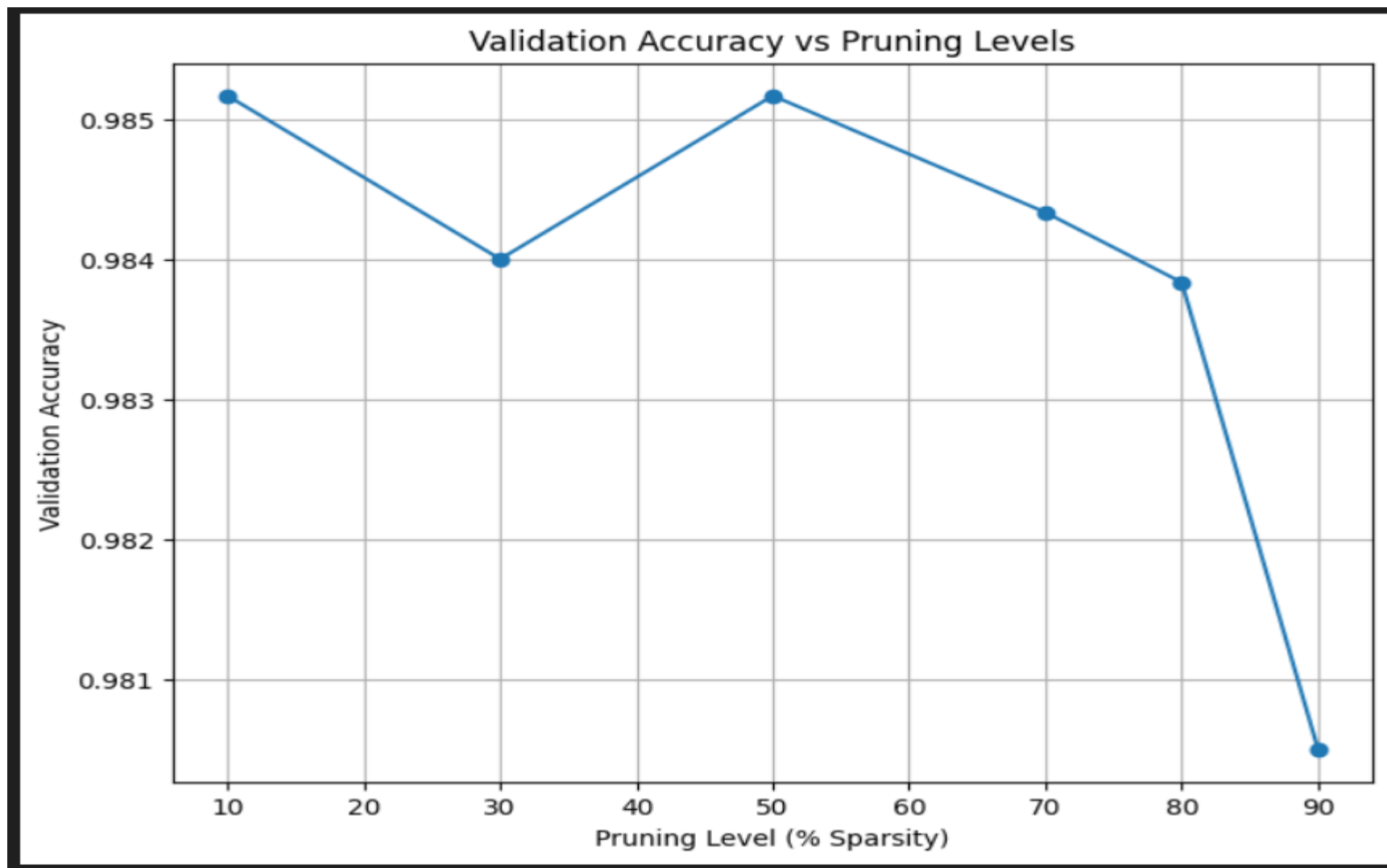
Magnitude-Based Pruning (accuracy)



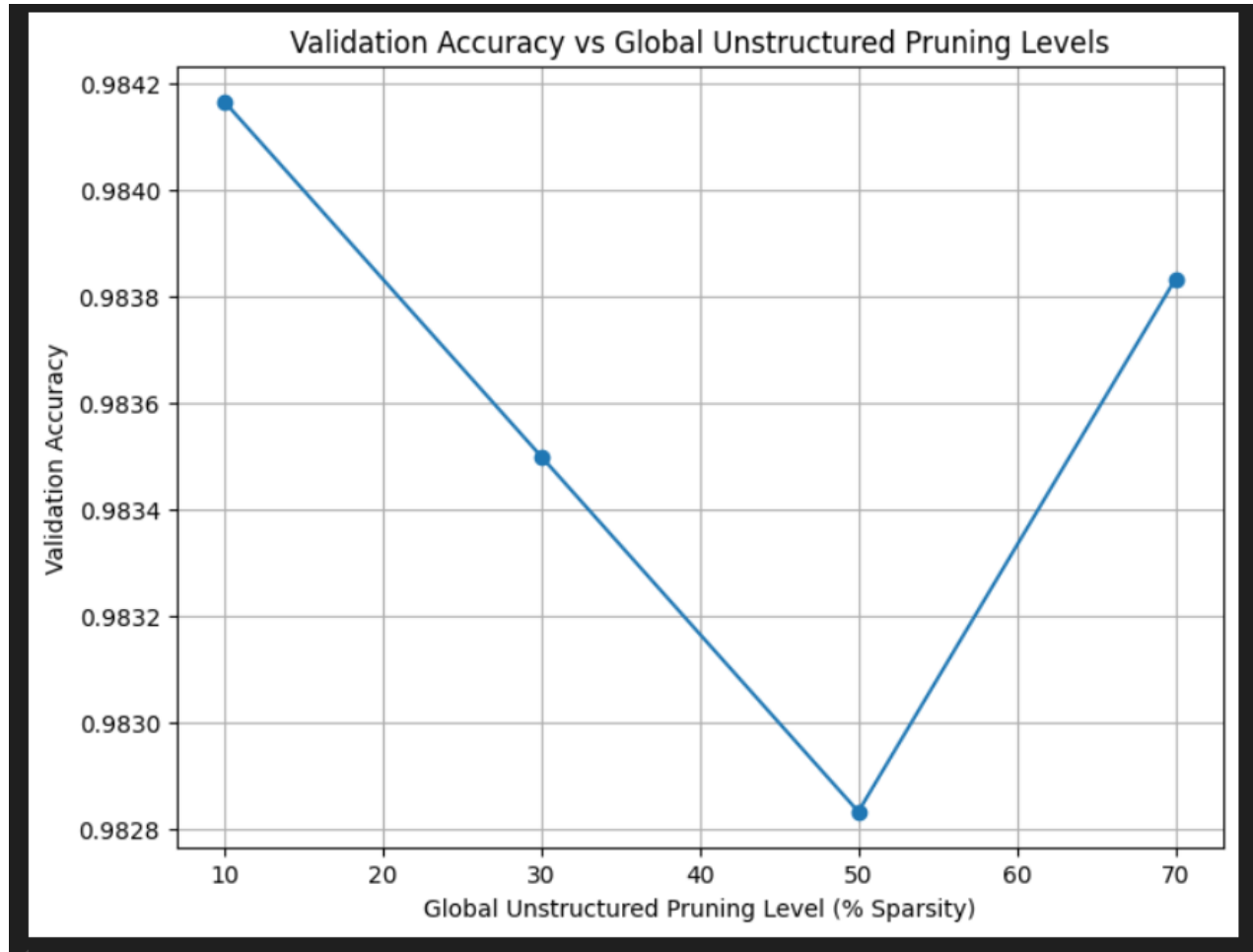
Structured Pruning on filters (accuracy)



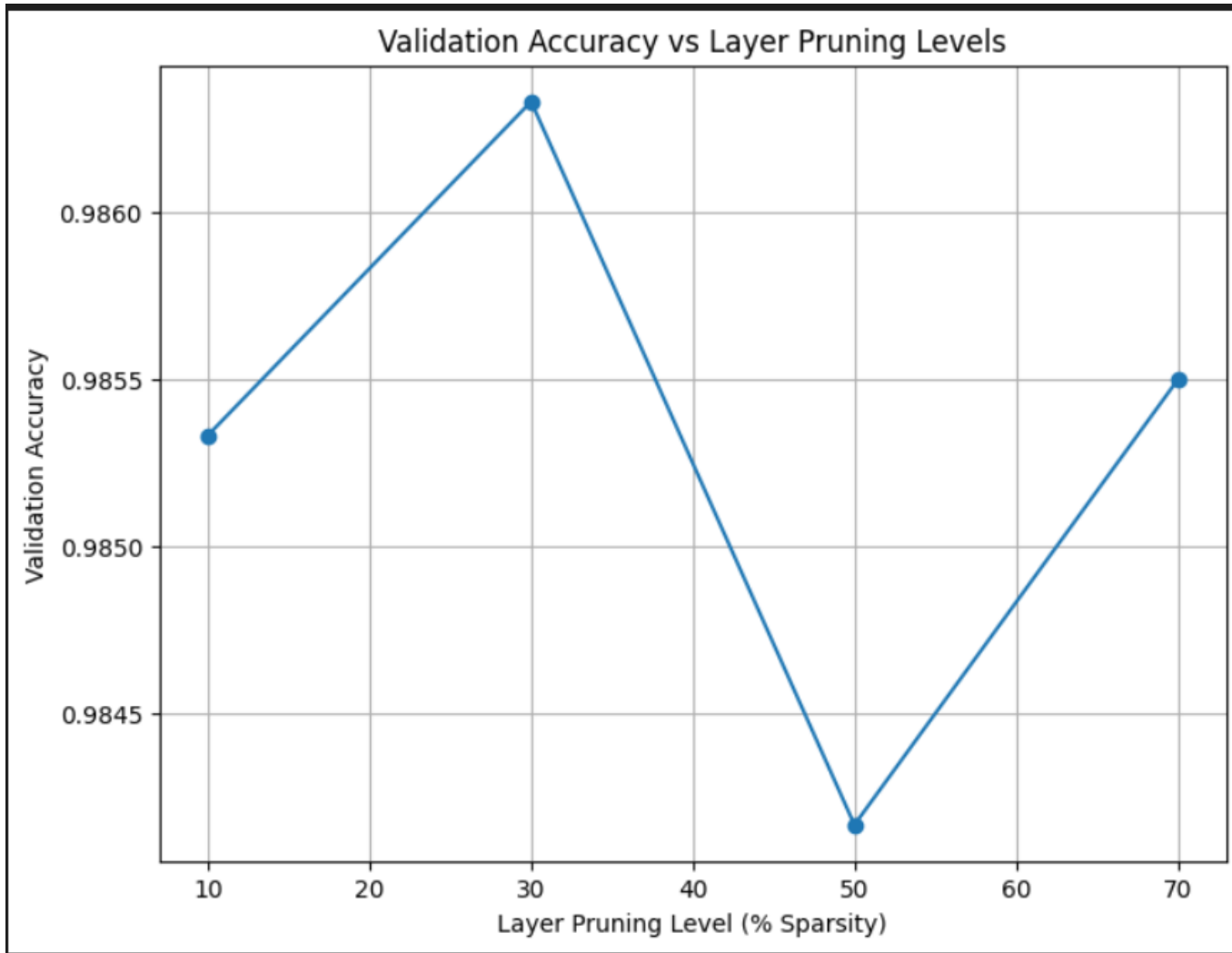
Structured Pruning on filters (accuracy)



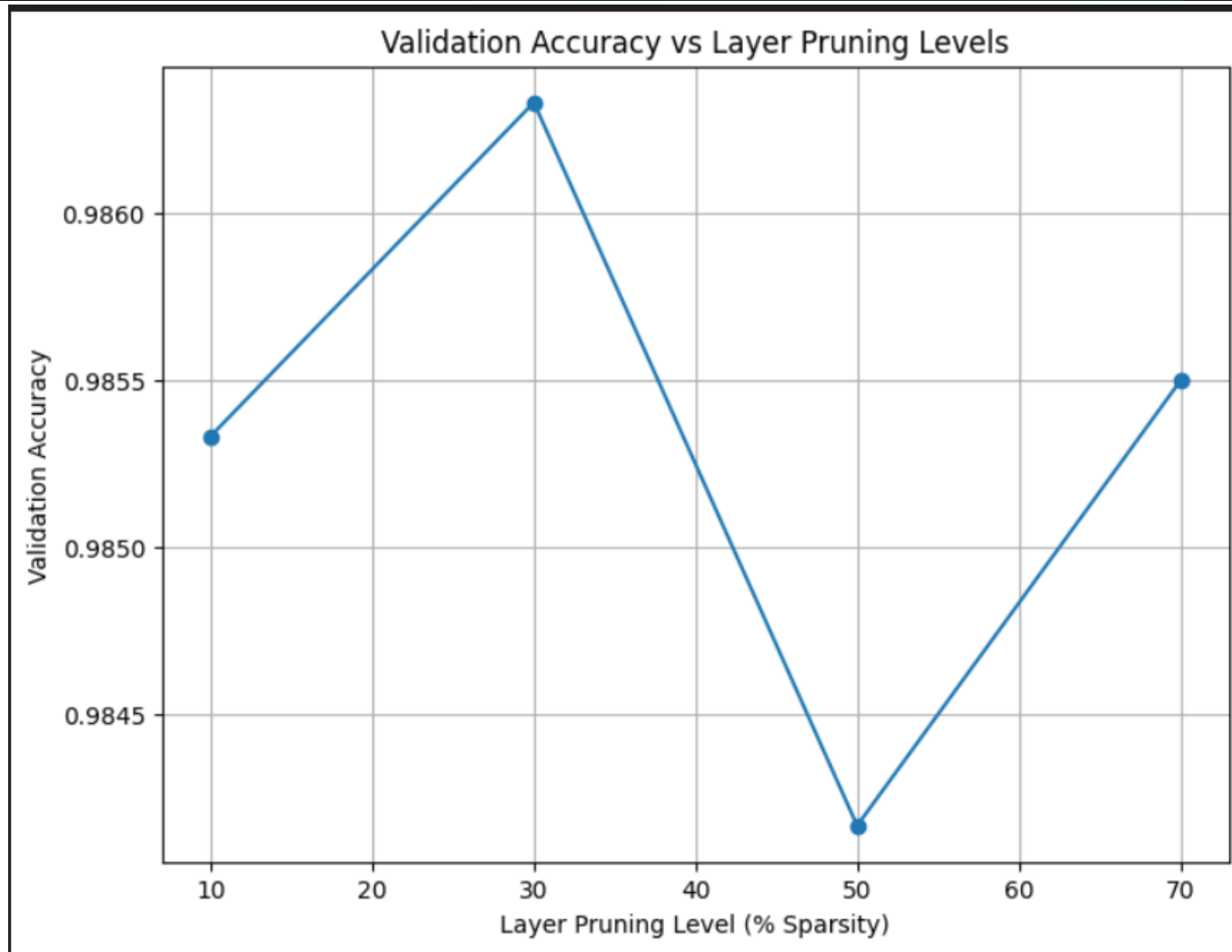
Unstructured Pruning Global (accuracy)



Prune layers (Conv2D and Dense) by individual weights



Prune entire layers(AutoML?) by Layer Level



Prune layer by importance (Conv2D more important)

